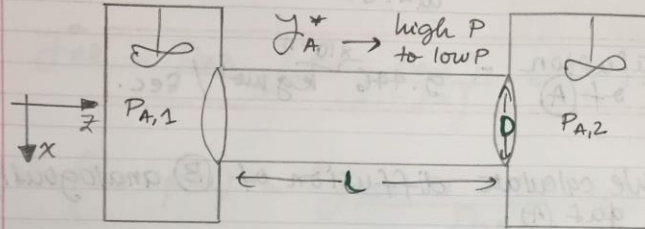


120/120

19.1-2 EQUIMOLAR COUNTERDIFFUSION of
 $N_3(A)$ and $N_2(B)$ at STEADY STATE



(a) Solution: Using Fick's Law

$$J_{Az}^* = -D_{AB} \frac{dC_A}{dz},$$

expressed in terms of partial pressures:

$$J_{Az}^* = \frac{D_{AB}}{RT} \frac{(P_{A1} - P_{A2})}{(z_2 - z_1)}, \text{ but}$$

$$\text{Diffusion of A} = (J_{Az}^*) \times (A_{\text{cylinder}}) = (\text{Flux}) \times (\text{Area})$$

$$= \frac{D_{AB} (P_{A1} - P_{A2})}{RT (z_2 - z_1)} (\pi r^2)$$

$$= \frac{(2.30 \times 10^{-5} \text{ m}^2/\text{s}) (20 \text{ kPa} - 6.667 \text{ kPa})}{(8314 \frac{\text{m}^3 \text{ Pa}}{\text{kgmol K}}) (298 \text{ K}) (0.610 \text{ m})} \times \pi \left(\frac{24.4}{2} \text{ mm} \right)^2$$

$$= \left(\frac{0.3067 \text{ m}^2 \text{ Pa} / \text{s}}{1511318.92 \text{ m}^2 \text{ Pa} / \text{kgmol}} \right) \times \pi (0.0122 \text{ m})^2$$

$$= (2.02 \times 10^{-7} \frac{\text{kg mol}}{\text{m}^2 \cdot \text{s}}) (4.676 \times 10^{-4} \text{ m}^2) \Rightarrow$$

$$\Rightarrow \text{Diffusion of (A)} = 9.446 \times 10^{-11} \text{ kg mol / sec.}$$

(b) We calculate diffusion of (B) analogously to gas (A):

$$J_{Bz}^* = N_B \times \text{Area}$$

$$N_B = -N_A = -2.029 \times 10^{-7} \text{ kg mol / m}^2 \cdot \text{s}$$

$$J_{Bz}^* = -D_{AB} \left(\frac{P_{A1} - P_{A2}}{r(21-22)} \right) (\pi r^2) = -J_{Az}^* \quad \left. \begin{array}{l} \\ y= \end{array} \right\}$$

$$-D_{AB} \equiv D_{BA}$$

$$\Rightarrow J_{Bz}^* = -J_{Az}^*$$

$$\Rightarrow J_{Bz}^* = -9.45 \times 10^{-11} \text{ kg mol / sec}$$

(c) Using the diffusivity equation, we solve for partial pressure:

$$J_{A2}^* = D_{AB} \frac{(P_{A1} - P_{A2})}{RT(z_1 - z_2)} \Rightarrow$$

$$\Rightarrow (P_{A1} - P_{A2}) = \frac{J_{A2}^* \cdot RT \cdot (z_1 - z_2)}{D_{AB}}$$

$$\Rightarrow P_{A1} = \frac{J_{A2}^* \cdot RT \cdot (z_1 - z_2)}{D_{AB}} + P_{A2}$$

$$= \frac{(2.029 \times 10^{-7} \frac{\text{kgmol}}{\text{m}^2 \cdot \text{s}}) (8314 \frac{\text{J}}{\text{kgmol} \cdot \text{K}}) (298 \text{ K}) (0.305 \text{ m})}{(2.30 \times 10^{-5} \text{ m}^2/\text{s})}$$

$$+ (6.667 \times 10^3 \text{ Pa})$$

$$= \frac{0.1533 \text{ (Pa/m}^2\text{s)}}{2.30 \times 10^{-5} \text{ (m}^2/\text{s)}} + (6.667 \times 10^3 \text{ Pa})$$

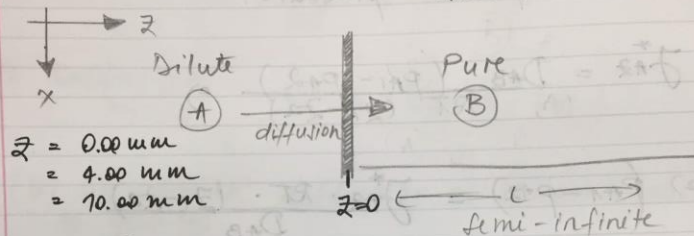
$$= (6.667 \times 10^3 \text{ Pa}) + (6.667 \times 10^3 \text{ Pa})$$

$$= 1.333 \times 10^4 \text{ Pa}$$

Thus,

$$P_{A1} = 1.333 \times 10^4 \text{ Pa}$$

19.1-17 Unsteady-State Diffusion and Reaction



At $z=0$, (A) enters in contact with membrane which contains a pure (B) medium. Then :

$$C_{A0} = 1.0 \text{ kg mol/m}^2 \text{ at } z=0.00 \text{ m.}$$

Using the (18.4-17) eqn, BUT WITH RXN term:

$$\frac{\partial C_A}{\partial t} = D_{AB} \left(\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right) - k' C_A$$

$$\frac{\partial C_A}{\partial t} = D_{AB} \left(\frac{\partial^2 C_A}{\partial z^2} \right) - k' C_A$$

Since we have a second-order PDE, we need 2 boundary conditions at z , and since we have an additional first order PDE wrt time, we need an initial condition for t .

I.C. at $t=0$, $C_A = 0$ for $z > 0$

B.C. at $z=0$, $C_A = C_{A0}$ for $t > 0$

B.C. at $z=\infty$, $C_A = 0$ for $t \gg 0$

Making use of the (20.2-5) eqn in the textbook, we know that the solution is:

$$\frac{C_A}{C_{A0}} = \frac{1}{2} \exp\left(-z \sqrt{\frac{k'}{D_{AB}}}\right) \cdot \operatorname{erfc}\left(\frac{z}{2\sqrt{t D_{AB}}} - \sqrt{k' t}\right) + \\ + \frac{1}{2} \exp\left(z \sqrt{\frac{k'}{D_{AB}}}\right) \cdot \operatorname{erfc}\left(\frac{z}{2\sqrt{t D_{AB}}} + \sqrt{k' t}\right).$$

(i) C_A at $z=0$

$$C_A = \left[\frac{1}{2} \exp(0) \cdot \operatorname{erfc}(0 - \sqrt{k' t}) + \right. \\ \left. + \frac{1}{2} \exp(0) \cdot \operatorname{erfc}(0 + \sqrt{k' t}) \right] \cdot C_{A0} \\ = \left[\frac{1}{2} \operatorname{erfc}(-\sqrt{k' t}) + \frac{1}{2} \operatorname{erfc}(\sqrt{k' t}) \right] C_{A0} \\ = \left[\frac{1}{2} \operatorname{erfc}\left(-\sqrt{(1 \times 10^{-4} \text{ s})(1 \times 10^5 \text{ s})}\right) + \right. \\ \left. + \frac{1}{2} \operatorname{erfc}\left(+\sqrt{(1 \times 10^{-4} \text{ s})(1 \times 10^5 \text{ s})}\right) \right] C_{A0}$$

$$\begin{aligned}
 \Rightarrow C_A &= \left(\frac{1}{2} \times \operatorname{erfc}(-\sqrt{10}) + \frac{1}{2} \operatorname{erfc}(\sqrt{10}) \right) C_{A0} \\
 &= \left(\frac{1}{2} \operatorname{erfc}(-3.162) + \frac{1}{2} \operatorname{erfc}(3.162) \right) \frac{1.0 \text{ kg mol}}{\text{m}^3} \\
 &= \left[\frac{1}{2} (1.999) + \frac{1}{2} (7.758 \times 10^{-6}) \right] C_{A0}
 \end{aligned}$$

$$\boxed{C_A|_{z=0} = 0.9995 \text{ kg mol/m}^3}$$

(ii) C_A at $z = 4.00 \text{ mm} = 0.004 \text{ m}$

$$\begin{aligned}
 C_A &= \left[\frac{1}{2} \exp\left(-0.004 \text{ m} \sqrt{\frac{1 \times 10^{-4} \text{ 1/s}}{1 \times 10^{-9} \text{ m}^2/\text{s}}}\right) \right. \\
 &\quad \left. + \operatorname{erfc}\left(\frac{0.004 \text{ m}}{2 \sqrt{(1 \times 10^5 \text{ s})(1 \times 10^{-9} \text{ m}^2/\text{s})}}\right) - \sqrt{\frac{(1 \times 10^{-4} \text{ 1/s})(1 \times 10^5 \text{ s})}{}} \right] \\
 &\quad + \frac{1}{2} \exp\left(0.004 \text{ m} \sqrt{\frac{1 \times 10^{-4} \text{ 1/s}}{1 \times 10^{-9} \text{ m}^2/\text{s}}}\right) \\
 &\quad \left. + \operatorname{erfc}\left(\frac{0.004 \text{ m}}{2 \sqrt{(1 \times 10^5 \text{ s})(1 \times 10^{-9} \text{ m}^2/\text{s})}}\right) + \sqrt{\frac{(1 \times 10^{-4} \text{ 1/s})(1 \times 10^5 \text{ s})}{}} \right] \\
 &\quad \times C_{A0}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow C_A &= \left[0.14113 \cdot \operatorname{erfc}(-2.96228) + \right. \\
 &\quad \left. + 1.7714 \cdot \operatorname{erfc}(3.36228) \right] C_{A0} \\
 &= \left[(0.14113)(1.9999) + (1.7714)(1.9846 \times 10^{-6}) \right] \times C_{A0} \\
 &= (0.282) C_{A0} = (0.282)(1.0 \text{ kg mol/m}^2)
 \end{aligned}$$

Thus,

$$C_A|_{z=4 \text{ mm}} = 0.282 \text{ kg mol/m}^2$$

(iii) C_A at $z = 10 \text{ mm}$

$$\begin{aligned}
 C_A &= \left[\frac{1}{2} \exp\left(-0.01 \frac{10^{-4}}{10^{-9}}\right) \times \operatorname{erfc}\left(\frac{0.01}{2\sqrt{10^5 \cdot 10^{-9}}} - \sqrt{10^{-4} \cdot 10^5}\right) \right. \\
 &\quad \left. + \frac{1}{2} \exp\left(0.01 \frac{10^{-4}}{10^{-9}}\right) \times \operatorname{erfc}\left(\frac{0.01}{2\sqrt{10^5 \cdot 10^{-9}}} + \sqrt{10^{-4} \cdot 10^5}\right) \right] \times C_{A0}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow C_A &= \left[(0.0212) \times \operatorname{erfc}(-2.662) + (11.812) \operatorname{erfc}(3.662) \right] \times C_{A0} \\
 &= \left[(0.0212)(1.9998) + (11.812)(2.232 \times 10^{-7}) \right] \times C_{A0} \\
 &= (0.0424) C_{A0} = (0.0424)(1.0 \text{ kg mol/m}^2)
 \end{aligned}$$

Thus,

$$C_A|_{z=10 \text{ mm}} = 0.0424 \frac{\text{kg mol}}{\text{m}^2}$$

If we want to know the kg mol of (A) absorbed/ m^2 ,
we compute Q : (eqn. 20.2-6)

$$Q = c_{A0} \sqrt{\frac{D_{AB}}{k'}} \left[\left(k't + \frac{1}{2} \right) \operatorname{erfc} \sqrt{k't} + \sqrt{\frac{k't}{\pi}} e^{-k't} \right]$$

$$= \left(1.0 \frac{\text{kg mol}}{m^2} \right) \sqrt{\frac{10^{-9} m^2/s}{10^{-4} 1/s}} \left[\left((10^{-4} \times 10^5) + \frac{1}{2} \right) \operatorname{erfc} \sqrt{(10^{-4})(10^5)} + \right.$$

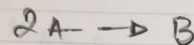
$$\left. + \sqrt{\frac{(10^{-4})(10^5)}{\pi}} e^{-10} \right] = (1.0) (\sqrt{10^{-5}}) \left[\left((10) + \frac{1}{2} \right) \operatorname{erfc} \sqrt{10} + \sqrt{\frac{10}{\pi}} e^{-10} \right]$$

$$\Rightarrow Q = 2.647876 \times 10^{-3} \frac{\text{kg mol (A)}}{m^2}$$

20/20

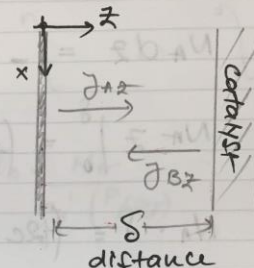
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19.1-21 Effect of a slow Reaction Rate on Diffusion



or

$$\begin{cases} N_A = -2N_B \\ N_B = -N_A/2 \end{cases}$$



Using (19.6-3):

$$N_A = -CD_{AB} \frac{dx_A}{dz} + x_A(N_A + N_B) \Rightarrow$$

where $N_B = -N_A/2$

$$\begin{aligned} \Rightarrow N_A &= -CD_{AB} \frac{dx_A}{dz} + x_A \left(\frac{2}{2} N_A + \left(-\frac{N_A}{2} \right) \right) \\ &= -CD_{AB} \frac{dx_A}{dz} + x_A \frac{N_A}{2} \end{aligned}$$

$$\Rightarrow N_A = \frac{x_A N_A}{2} = -CD_{AB} \frac{dx_A}{dz}$$

$$N_A \left(1 - \frac{x_A}{2} \right) = -CD_{AB} \frac{dx_A}{dz}$$

$$N_A \left(\frac{2 - x_A}{2} \right) = -CD_{AB} \frac{dx_A}{dz}$$

$$\Rightarrow N_A \left(\frac{2-x_A}{2} \right) dz = -c D_{AB} dx_A \quad \Big| \cdot \frac{2}{2-x_A}$$

$$\int N_A dz = \int - \frac{2c D_{AB}}{2-x_A} dx_A$$

$$N_A z \Big|_0^{\delta} = \left(+ 2c D_{AB} \right) \ln(2-x_A) \Big|_{x_{A1}}^{x_{A2}}$$

$$N_A \cdot \delta = \left(+ 2c D_{AB} \right) \ln \left(\frac{2-x_{A2}}{2-x_{A1}} \right) \quad | : \delta$$

$$N_A = + \frac{2c D_{AB}}{\delta} \ln \left(\frac{2-x_{A2}}{2-x_{A1}} \right) *$$

Very fast rxn $\Rightarrow x_{A2} = 0$

$$\Rightarrow N_A = + \frac{2c D_{AB}}{\delta} \ln \left(\frac{2}{2-x_{A1}} \right)$$

(b) Given the parameters, we solve for N_A :

$$N_A = \frac{2 \left(\frac{P}{RT} \right) (0.2 \times 10^{-4} \text{ m}^2/\text{s})}{(0.0013 \text{ m})} \ln \left(\frac{2}{2-0.97} \right)$$

$$= \frac{2 \left(\frac{101320 \text{ Pa}}{(8.314) (298)} \right) (0.2 \times 10^{-4})}{(0.0013)} \ln(1.9417)$$

$$N_A = 8.34965 \times 10^{-4} \text{ kgmol/m}^2 \cdot \text{s}$$

(c) Flow 1st order reaction at surface.
We use (19.6-7):

$$N_{A2-\delta} = k_1' c_{A2} \\ = k_1' c x_{A2}$$

$$\left. \begin{array}{l} \text{where } x_{A2} = \frac{N_A}{k_1' c} \\ c = \frac{P}{RT} \end{array} \right\} \Rightarrow x_{A2} = \frac{N_A}{k_1' (P/RT)}$$

From *, we get:

$$N_A = \frac{2 c D_{AB}}{\delta} \ln \left(\frac{2 - x_{A2}}{2 - x_{A1}} \right)$$

$$N_A = \frac{2 (P/RT) D_{AB}}{\delta} \ln \left(\frac{2 - N_A / k_1' (P/RT)}{2 - x_{A1}} \right)$$

$\Rightarrow$

$$N_A = \left(\frac{2 P D_{AB}}{\delta R T} \right) \ln \left(\frac{RT (2 - N_A) / k_1' P}{2 - x_{A1}} \right)$$

(d) Solving by trial and error, as suggested in example 19.6-1, we can get

$$N_A = 1.766 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s}$$

and then

$$x_{A2} = 0.8148$$

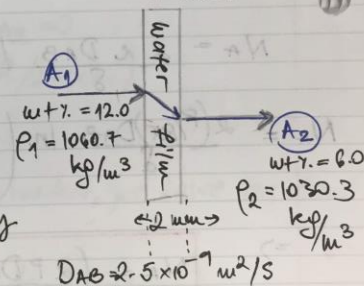
19.2-1 Diffusion of (A) through stagnant (B) in a liquid $T = 283 \text{ K}$

Assumption:

→ Steady State

→ water not diffused in region ①

⇓
water impermeable boundary



(Region ①) : $x_{A1} = \frac{\frac{\text{wt}\%}{M_A}}{\frac{\text{wt}\%}{M_A} + \frac{1-\text{wt}\%}{M_B}}$; $M_A = \text{molec. wt. of HCl}$
 $M_B = \text{molec. wt. of H}_2\text{O}$

$$\Rightarrow x_{A1} = \frac{\frac{12}{36.5}}{\frac{12}{36.5} + \frac{88}{18}} = 0.0629$$

Then $x_{B1} = 1.00 - x_{A1}$, Average molec. wt. :
 $= 1 - 0.0629$
 $= 0.9371$

$$\bar{M}_1 = \frac{100}{\frac{wt\% \cdot A_1}{M_A} + \frac{wt\% \cdot B_1}{M_B}}$$

$$\bar{M}_1 = \frac{100}{12/36.5 + 88/18} = 19.167 \text{ kg/kmol}$$

$x_{A1} = 0.0629$
 $x_{B1} = 0.9371$

Region :
 (2) $x_{A2} = \frac{wt\% \cdot A_2 / M_A}{\frac{wt\% \cdot A_2}{M_A} + \frac{1 - wt\% \cdot A_2}{M_B}}$

$\rightarrow x_{A2} = \frac{6/36.5}{6/36.5 + 94/18} = 0.0299 \approx 0.03$

Then $x_{B2} = 1.00 - x_{A2} = 1.00 - 0.03 = 0.97$

$x_{A2} = 0.03$
 $x_{B2} = 0.97$

$$\bar{M}_2 = \frac{100}{6/36.5 + 94/18} = 18.556 \text{ kg/molK}$$

Furthermore,

$$c_{avg} = \left(\frac{c}{M} \right)_{avg} = \frac{c_1/M_1 + c_2/M_2}{2}$$

$$= \frac{(1060.7/19.167) + (1030.3/18.556)}{2} = 55.43 \frac{\text{kmol}}{\text{m}^3}$$

and we need to know:

$$x_{Bm} = \frac{x_{B2} - x_{B1}}{\ln(x_{B2}/x_{B1})} = \frac{0.97 - 0.937}{\ln(0.97/0.937)}$$

$$\Rightarrow x_{Bm} = 0.956$$

Finally, we compute the flux for (A):

$$N_A = \frac{C D_{AB}}{x_{Bm} \cdot S} (x_{A1} - x_{A2})$$

$$= (55.43) \left(\frac{2.5 \times 10^{-9}}{(0.956)(0.002)} \right) (0.063 - 0.03)$$

$$\Rightarrow N_A = 2.372 \times 10^{-6} \text{ mol/kg/m}^2 \cdot \text{s}$$

19.3-1 Diffusion of CO₂ through Rubber

The equation for gas diffusion through solid is governed by:

$$N_A = \frac{D_{AB} \cdot S \cdot (P_{A1} - P_{A2})}{(z_2 - z_1)}$$

Since $c = \frac{P}{RT} \Rightarrow P_{A1} = c_{A1} \cdot RT$

$$P_{A2} = c_{A2} \cdot RT$$

recall that

$$\text{conc.} = A \cdot P \Rightarrow C_1 = S \cdot P_1 = (0.9 \text{ m}^3)(2 \text{ atm})$$

$$\Rightarrow C_1 = (1.8) / (22.414) \text{ m}^3/\text{kg mol} \Rightarrow$$

$$\Rightarrow C_1 = 0.082 \text{ kg mol}/\text{m}^3$$

but $c_2 = 0$, since $P_2 = 0$;

going back to:

$$N_A = \frac{D_{AB} (C_{A1} - C_{A2}) \cdot A}{(z_2 - z_1)}$$

$$= \frac{(2.5 \times 10^{-9} \text{ m}^2/\text{s})(0.082) \text{ kg mol}/\text{m}^3 (4.0 \times 10^{-4} \text{ m}^2)}{0.03 \text{ m}}$$

$$\Rightarrow N_A = 1.176 \times 10^{-13} \text{ kg mol}/\text{s} \cdot \text{m}^2 \checkmark$$

19.3-6 Diffusion of CO_2 in a Packed Bed of Sand.

ASSUMPTIONS:

- diffusion of CO_2 at steady-state
- diffusion through a packed bed of sand
- no convection

$$T = 276 \text{ K}$$

$$P = 1.013 \times 10^5 \text{ Pa}$$

$$L = 1.25 \text{ m}$$

$$\epsilon = 0.3$$

$$P_{A,1} = 2.026 \times 10^3$$

$$D_{AB} =$$

$$P_{A,2} = 0 \text{ Pa}$$

$$T = 1.87$$

The rate of diffusion of CO_2 is given by the expression:

$$N_A = \frac{D_{AB} \epsilon (P_{A,1} - P_{A,2})}{RT^2 (z_2 - z_1)}$$

Being given all parameters, we calculate:

$$N_A = \frac{(0.142 \times 10^{-4}) (0.30) (2.026 \times 10^3)}{(8.314) (276) (1.87) (1.25)}$$

$$= \frac{0.4345}{5.338 \times 10^3} = \frac{8.633 \times 10^{-3}}{5.338 \times 10^3} = 1.616 \times 10^{-6} \frac{\text{mol}}{\text{m}^2 \text{ s}}$$

Thus, $N_A = 1.616 \times 10^{-6} \text{ mol / m}^2 \cdot \text{s}$

20/20